

LAB EXPERIMENTS

Tool: Python & R Programming

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Subject: Data Handling & Visualization

Code: DSA0604

Set 1: Monthly Sales Data

1. Create a Line Chart to Visualize Monthly Sales

CODE:

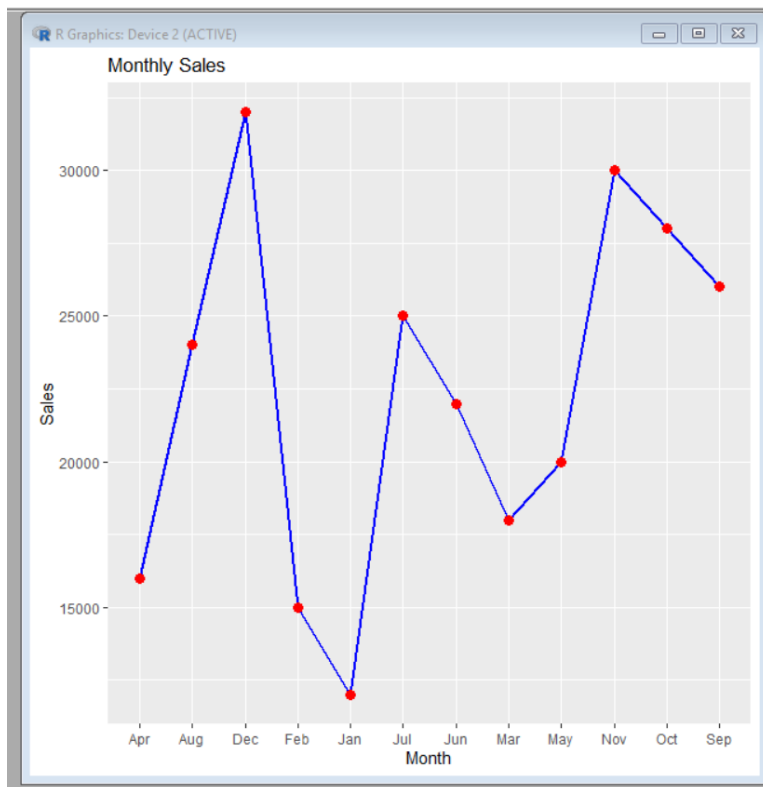
```
library(ggplot2)

sales_data <- data.frame(

  Month = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
  Sales = c(12000, 15000, 18000, 16000, 20000, 22000,
            25000, 24000, 26000, 28000, 30000, 32000)
)

ggplot(sales_data, aes(x=Month, y=Sales, group=1)) +
  geom_line(color="blue", linewidth=1) +
  geom_point(color="red", size=3) +
  labs(title="Monthly Sales",
       x="Month",
       y="Sales")
```

OUTPUT:



2. Generate a Bar Chart for Top-Selling Products

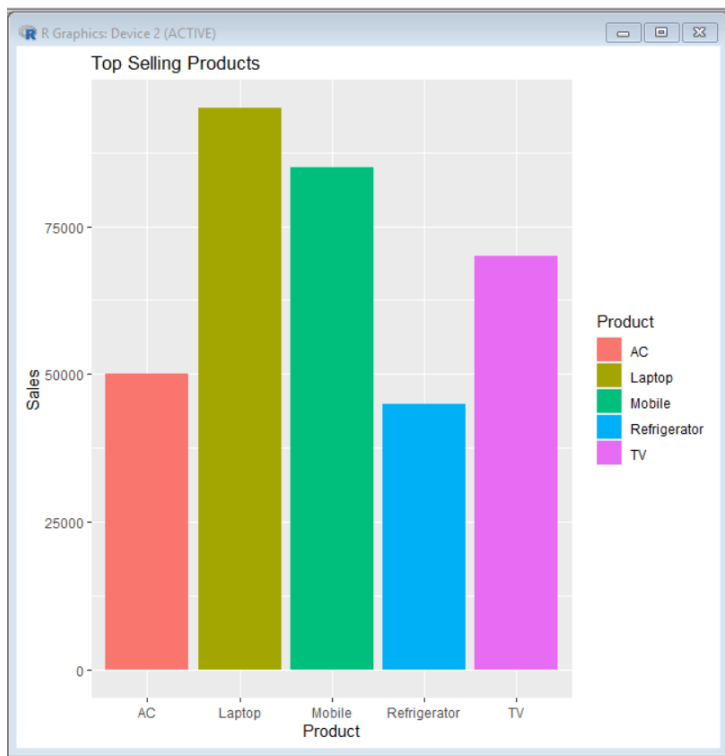
CODE:

```
library(ggplot2)

product_data <- data.frame(
  Product = c("Laptop", "Mobile", "TV", "AC", "Refrigerator"),
  Sales = c(95000, 85000, 70000, 50000, 45000)
)

ggplot(product_data,
  aes(x=Product, y=Sales, fill=Product)) +
  geom_bar(stat="identity") +
  labs(title="Top Selling Products",
    x="Product",
    y="Sales")
```

OUTPUT:



3. Scatter Plot: Advertising Budget vs Monthly Sales

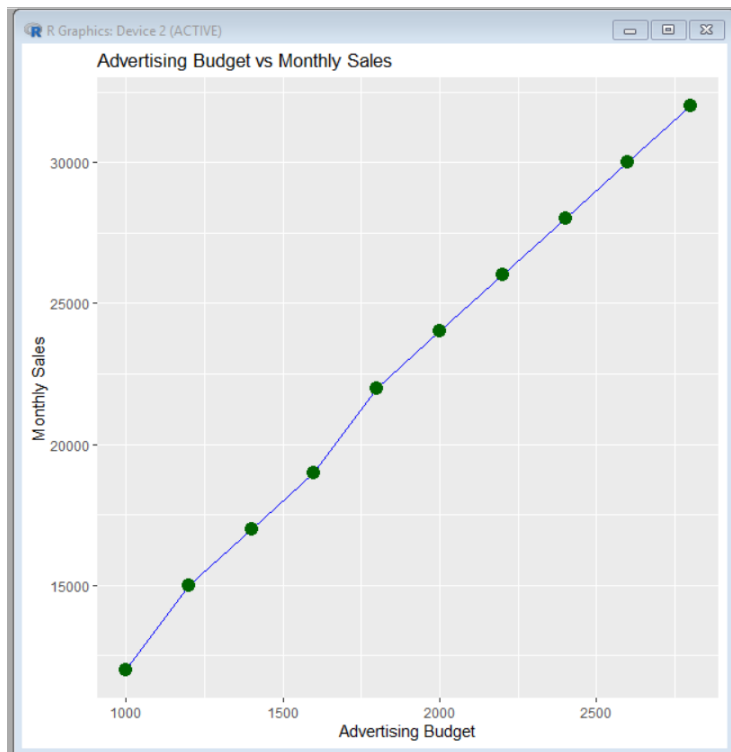
CODE:

```
library(ggplot2)

sales_data <- data.frame(
  AdBudget = c(1000,1200,1400,1600,1800,2000,2200,2400,2600,2800),
  Sales = c(12000,15000,17000,19000,22000,24000,26000,28000,30000,32000)
)

ggplot(sales_data, aes(x=AdBudget, y=Sales)) +
  geom_line(color="blue") +
  geom_point(color="darkgreen", size=4) +
  labs(title="Advertising Budget vs Monthly Sales",
       x="Advertising Budget",
       y="Monthly Sales")
```

OUTPUT:



4. Interactive Dashboard (Line Chart + Bar Chart)

CODE:

```
library(plotly)

sales_data <- data.frame(
  Month = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
  Sales = c(12000, 15000, 18000, 16000, 20000, 22000,
            25000, 24000, 26000, 28000, 30000, 32000)
)

product_data <- data.frame(
  Product = c("Laptop", "Mobile", "TV", "AC", "Refrigerator"),
  Sales = c(95000, 85000, 70000, 50000, 45000)
)

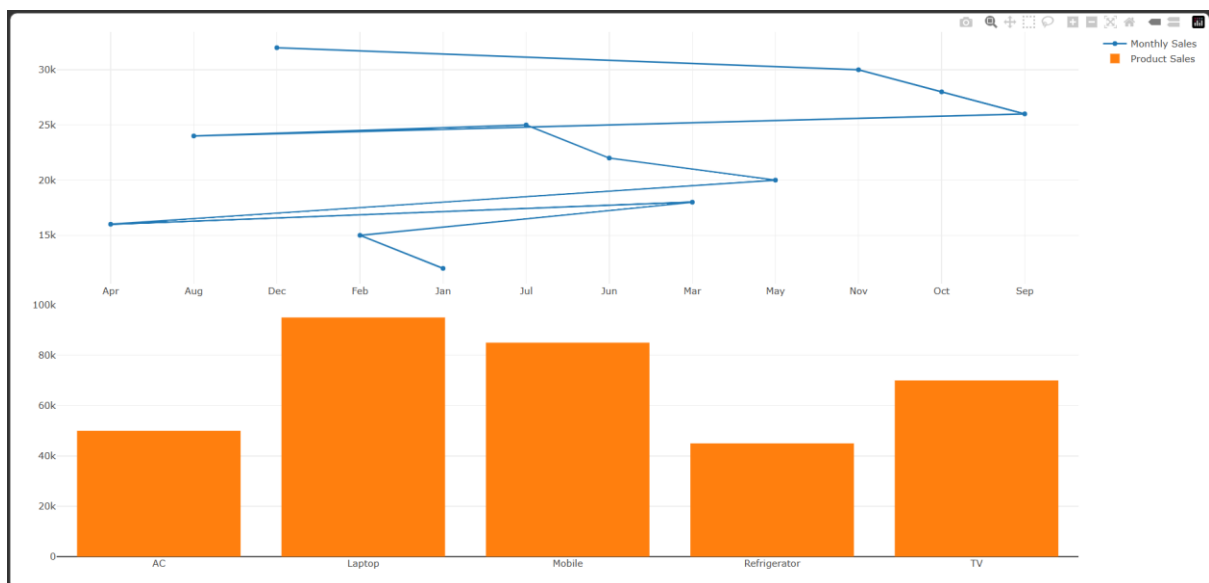
line_plot <- plot_ly(
  sales_data,
  x = ~Month,
  y = ~Sales,
```

```

type = "scatter",
mode = "lines+markers",
name = "Monthly Sales"
)
bar_plot <- plot_ly(
  product_data,
  x = ~Product,
  y = ~Sales,
  type = "bar",
  name = "Product Sales"
)
subplot(line_plot, bar_plot, nrow = 2)

```

OUTPUT:



Set 2: Customer Feedback Analysis.

1. Create a Histogram for Customer Ages

CODE:

```

library(ggplot2)
customer <- data.frame(
  CustomerID = c(1,2,3,4,5),

```

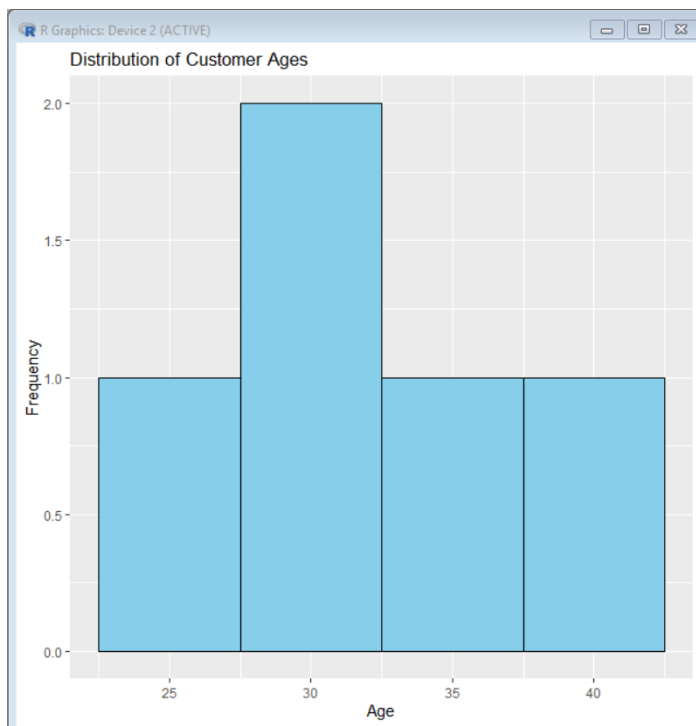
```

Age = c(25,30,35,28,40),
Satisfaction = c(4,5,3,4,5)
)

ggplot(customer, aes(x=Age)) +
  geom_histogram(binwidth=5, fill="skyblue", color="black") +
  labs(title="Distribution of Customer Ages",
        x="Age",
        y="Frequency")

```

OUTPUT:



2. Create a Pie Chart for Satisfaction Scores

CODE:

```

customer <- data.frame(
  Satisfaction = c(4,5,3,4,5)
)

score <- table(customer$Satisfaction)

pie(score,
     labels=paste(names(score), score),

```

```
col=rainbow(length(score)),  
main="Customer Satisfaction Scores")
```

OUTPUT:

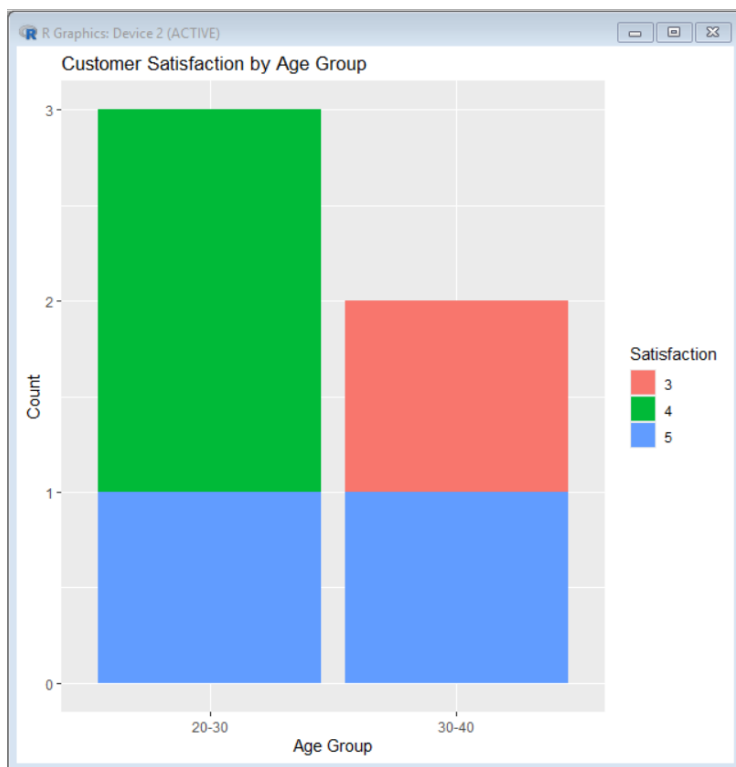


3. Create a Stacked Bar Chart by Age Group

CODE:

```
library(ggplot2)  
customer <- data.frame(  
  AgeGroup = c("20-30", "20-30", "30-40", "20-30", "30-40"),  
  Satisfaction = c("4", "5", "3", "4", "5")  
)  
ggplot(customer,  
  aes(x=AgeGroup, fill=Satisfaction)) +  
  geom_bar() +  
  labs(title="Customer Satisfaction by Age Group",  
    x="Age Group",  
    y="Count")
```

OUTPUT:



4. Create a Word Cloud

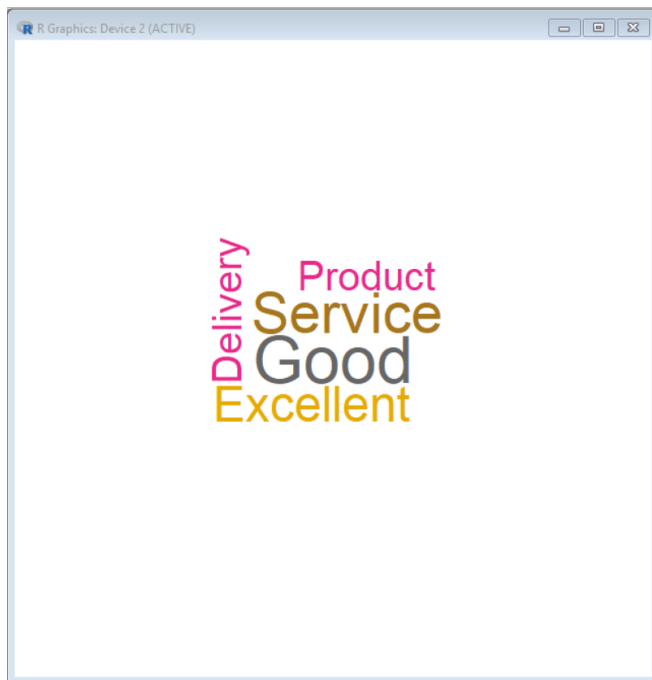
CODE:

```
library(wordcloud)
library(RColorBrewer)
words <- c("Good", "Service", "Excellent", "Product",
           "Delivery", "Quality", "Support",
           "Happy", "Customer", "Fast")

freq <- c(6, 5, 4, 3, 3, 2, 2, 2, 1, 1)

wordcloud(
  words = words,
  freq = freq,
  scale = c(4, 1),
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2")
)
```


OUTPUT:



Set 3: Employee Performance Evaluation.

1.Create a Line Chart to Visualize Employee Performance Trend

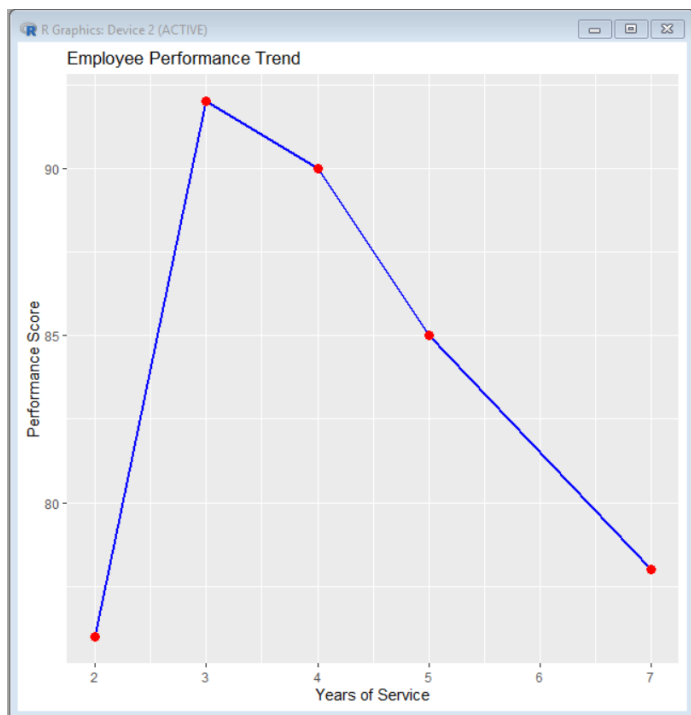
CODE:

```
library(ggplot2)

employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsService = c(5,3,7,4,2),
  Performance = c(85,92,78,90,76)
)

ggplot(employee, aes(x=YearsService, y=Performance, group=1)) +
  geom_line(color="blue", linewidth=1) +
  geom_point(color="red", size=3) +
  labs(title="Employee Performance Trend",
       x="Years of Service",
       y="Performance Score")
```

OUTPUT:



2. Generate a Bar Chart Showing Employees in Each Department

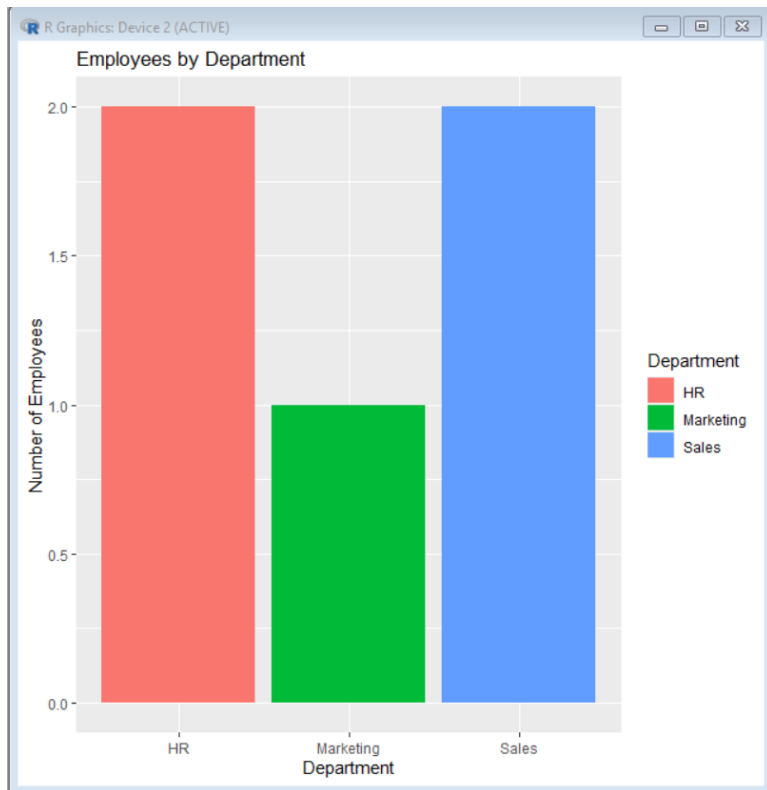
CODE:

```
library(ggplot2)

employee <- data.frame(
  Department = c("Sales", "HR", "Marketing", "Sales", "HR")
)

ggplot(employee, aes(x=Department, fill=Department)) +
  geom_bar() +
  labs(title="Employees by Department",
       x="Department",
       y="Number of Employees")
```

OUTPUT:



3. Scatter Plot: Years of Service vs Performance Score

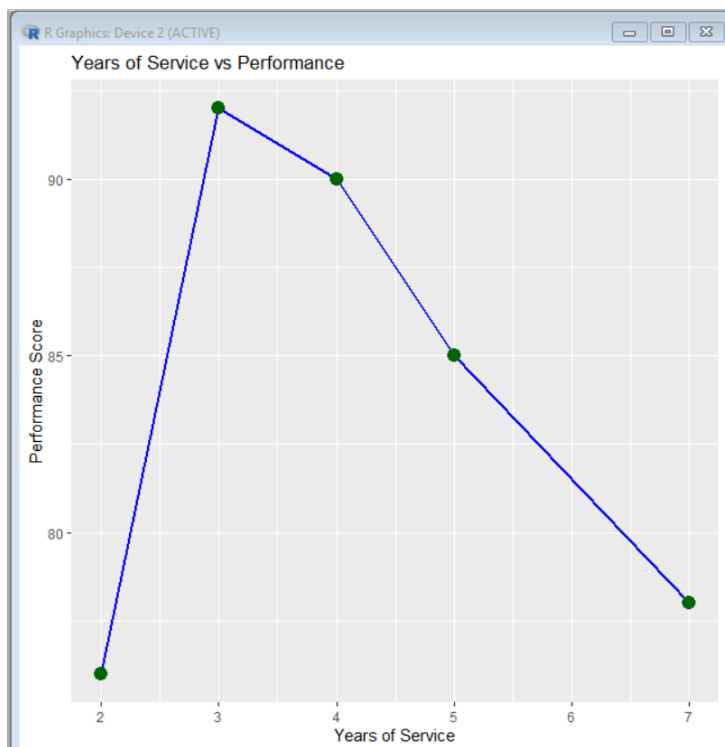
CODE:

```
library(ggplot2)

employee <- data.frame(
  YearsService = c(2,3,4,5,7),
  Performance = c(76,92,90,85,78)
)

ggplot(employee, aes(x = YearsService,
  y = Performance,
  group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "darkgreen", size = 4) +
  labs(
    title = "Years of Service vs Performance",
    x = "Years of Service",
    y = "Performance Score")
```

OUTPUT:



4. Interactive Dashboard (Line Chart + Bar Chart)

CODE:

```
library(ggplot2)
library(patchwork)

employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsService = c(5,3,7,4,2),
  Performance = c(85,92,78,90,76)
)

employee <- employee[order(employee$YearsService), ]

p1 <- ggplot(employee,
  aes(x=YearsService,
    y=Performance,
    group=1)) +
  geom_line(color="blue", linewidth=1.2) +
```

```

geom_point(color="red", size=3) +
labs(title="Employee Performance Trend",
      x="Years of Service",
      y="Performance Score") +
theme_minimal()
dept <- as.data.frame(table(employee$Department))
colnames(dept) <- c("Department","Count")
p2 <- ggplot(dept,
             aes(x=Department,
                 y=Count,
                 fill=Department)) +
geom_col() +
labs(title="Employees by Department",
      x="Department",
      y="Employees") +
theme_minimal()
p1 + p2

```

OUTPUT:



Set 4: Product Inventory Management

1.Create a Bar Chart to Visualize Product Quantity

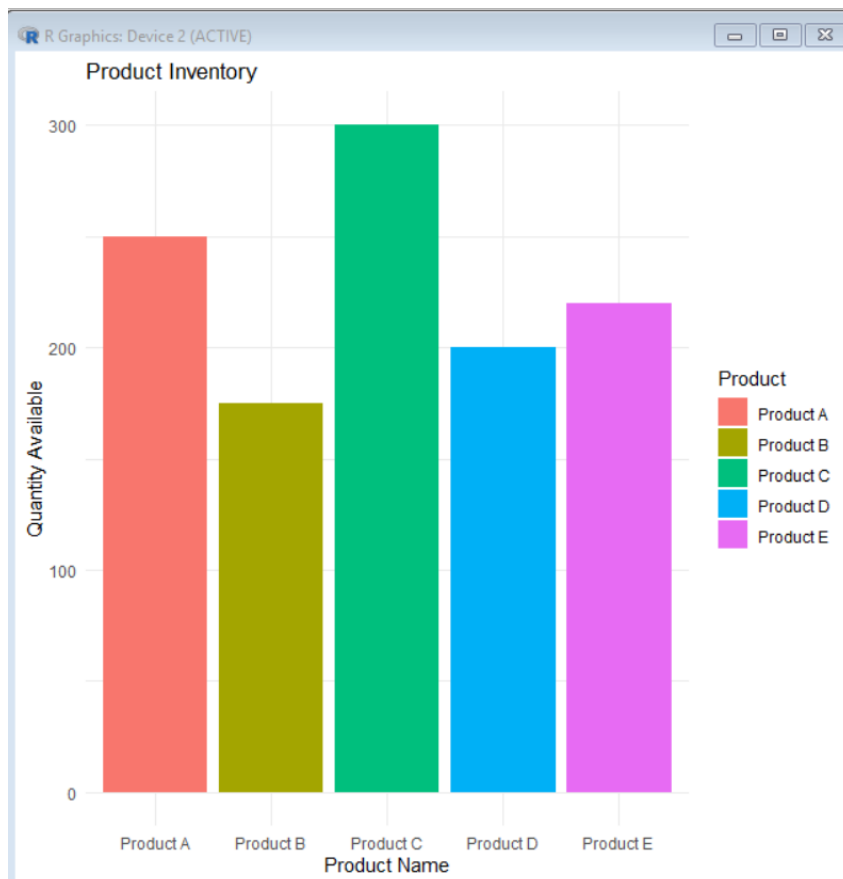
CODE:

```
library(ggplot2)

inventory <- data.frame(
  Product = c("Product A","Product B","Product C","Product D","Product E"),
  Quantity = c(250,175,300,200,220)
)

ggplot(inventory, aes(x=Product, y=Quantity, fill=Product)) +
  geom_bar(stat="identity") +
  labs(title="Product Inventory",
       x="Product Name",
       y="Quantity Available") +
  theme_minimal()
```

OUTPUT:



2. Generate a Stacked Bar Chart

CODE:

```
library(ggplot2)

inventory <- data.frame(

  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Stationery"),
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220)
)

ggplot(inventory,
  aes(x=Category,
      y=Quantity,
      fill=Product)) +
  geom_bar(stat="identity") +
  labs(title="Product Quantity by Category",
       x="Category",
       y="Quantity") +
  theme_minimal()
```

OUTPUT:



3. Scatter Plot: Product Price vs Quantity Available

CODE:

```
library(ggplot2)

inventory <- data.frame(

  Product = c("Product A","Product B","Product C","Product D","Product E"),

  Price = c(500,700,600,800,650),

  Quantity = c(250,175,300,200,220)

)

inventory <- inventory[order(inventory$Price), ]

ggplot(inventory,

  aes(x = Price,

    y = Quantity,

    group = 1)) +

  geom_line(color = "blue", linewidth = 1) +

  geom_point(color = "purple", size = 4) +

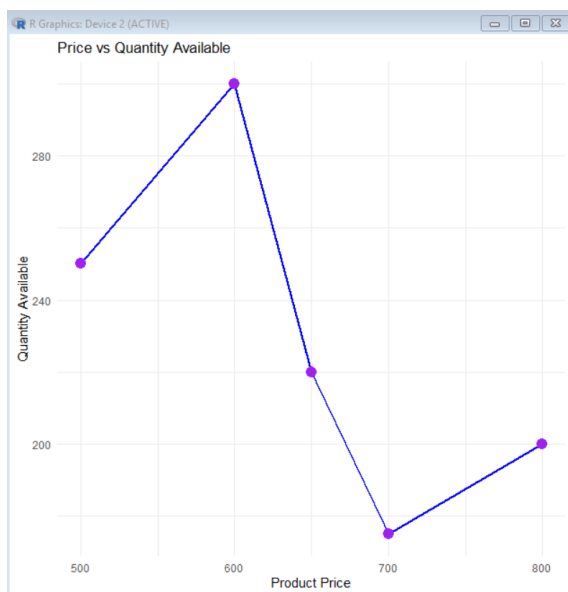
  labs(title = "Price vs Quantity Available",

    x = "Product Price",

    y = "Quantity Available") +

  theme_minimal()
```

OUTPUT:



4. Dashboard (Bar Chart + Stacked Bar Chart)

CODE:

```
library(ggplot2)
library(patchwork)

inventory1 <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220)
)

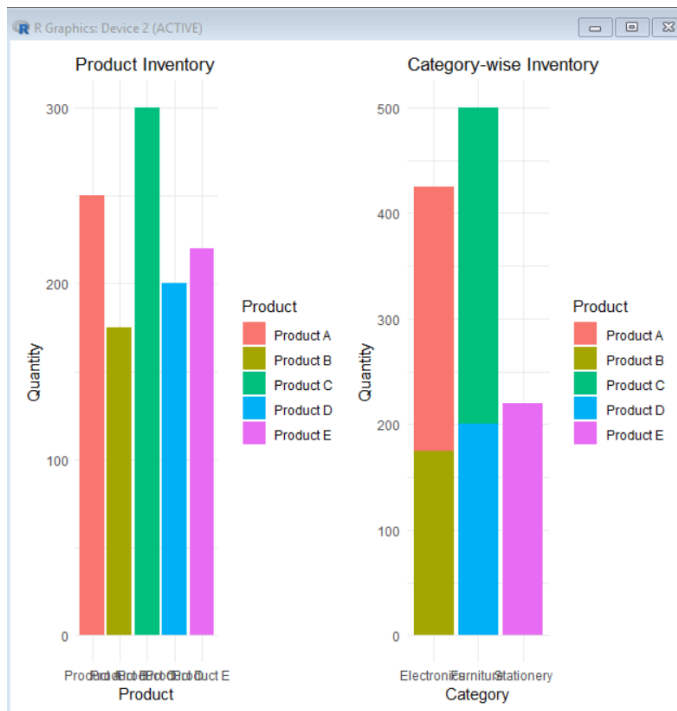
inventory2 <- data.frame(
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Stationery"),
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220)
)

p1 <- ggplot(inventory1,
  aes(Product, Quantity, fill=Product)) +
  geom_bar(stat="identity") +
  labs(title="Product Inventory") +
  theme_minimal()

p2 <- ggplot(inventory2,
  aes(Category, Quantity, fill=Product)) +
  geom_bar(stat="identity") +
  labs(title="Category-wise Inventory") +
  theme_minimal()

p1 + p2
```

OUTPUT:



Set 5: Website Analytics

1.Create a Line Chart for Daily Page Views

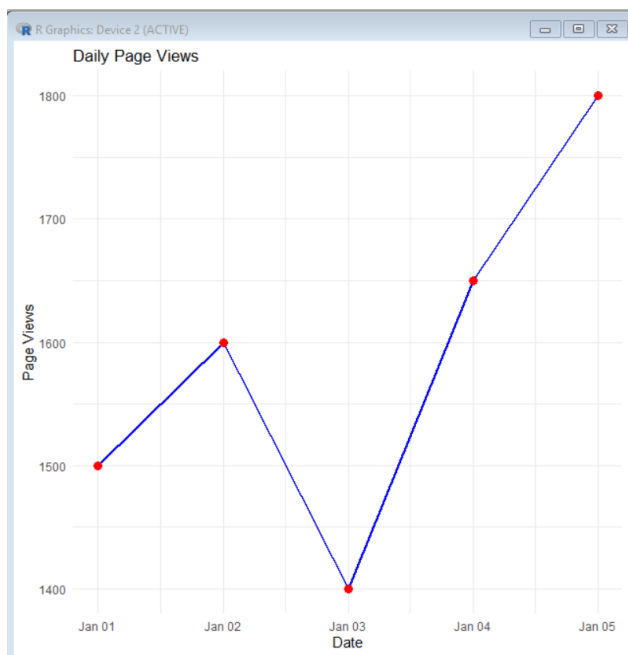
CODE:

```
library(ggplot2)

website <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
    "2023-01-04", "2023-01-05")),
  PageViews = c(1500, 1600, 1400, 1650, 1800)
)

ggplot(website, aes(x=Date, y=PageViews, group=1)) +
  geom_line(color="blue", linewidth=1) +
  geom_point(color="red", size=3) +
  labs(title="Daily Page Views",
    x="Date",
    y="Page Views") +
  theme_minimal()
```

OUTPUT:



2. Generate a Bar Chart for Click-Through Rate

CODE:

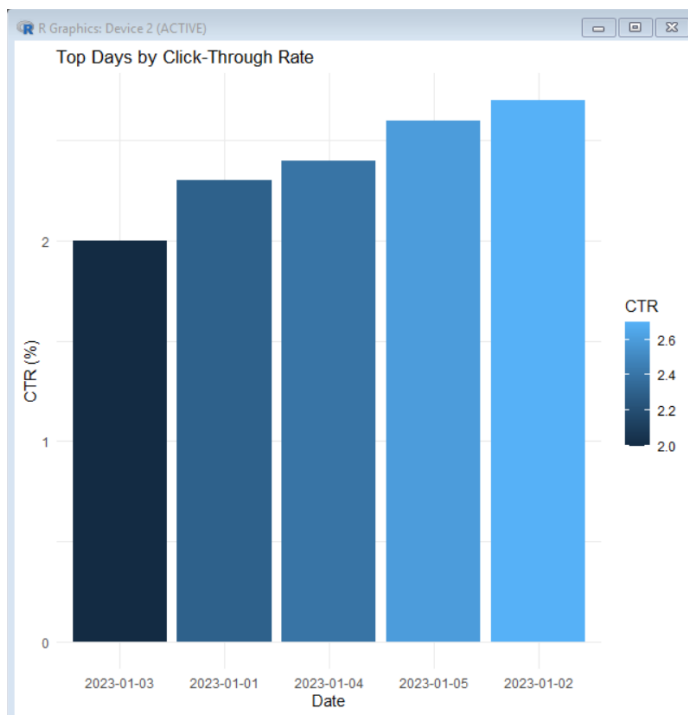
```
library(ggplot2)

website <- data.frame(
  Date = c("2023-01-01", "2023-01-02", "2023-01-03",
           "2023-01-04", "2023-01-05"),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

website <- website[order(-website$CTR), ]

ggplot(website,
  aes(x=reorder(Date, CTR), y=CTR, fill=CTR)) +
  geom_bar(stat="identity") +
  labs(title="Top Days by Click-Through Rate",
       x="Date",
       y="CTR (%)") +
  theme_minimal()
```

OUTPUT:



3. Create a Stacked Area Chart

CODE:

```
library(ggplot2)

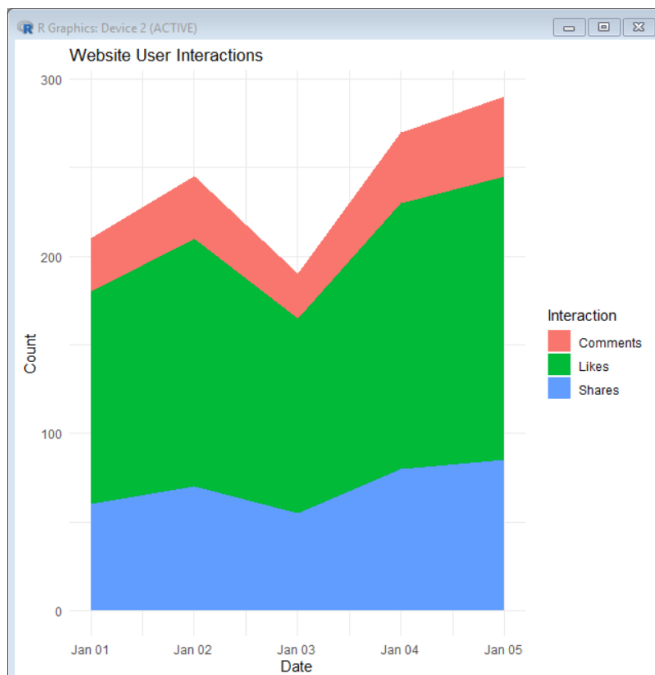
website <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
    "2023-01-04", "2023-01-05")),
  Likes = c(120, 140, 110, 150, 160),
  Shares = c(60, 70, 55, 80, 85),
  Comments = c(30, 35, 25, 40, 45)
)

long_data <- data.frame(
  Date = rep(website$Date, 3),
  Interaction = rep(c("Likes", "Shares", "Comments"),
    each=nrow(website)),
  Count = c(website$Likes,
    website$Shares,
    website$Comments)
```

```
)

ggplot(long_data,
       aes(x=Date,
           y=Count,
           fill=Interaction)) +
  geom_area() +
  labs(title="Website User Interactions",
       x="Date",
       y="Count") +
  theme_minimal()
```

OUTPUT:



4. Dashboard (Line Chart + Bar Chart)

CODE:

```
library(ggplot2)

library(patchwork)

website1 <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05")),
```

```

PageViews = c(1500,1600,1400,1650,1800)
)
website2 <- data.frame(
  Date = c("2023-01-01","2023-01-02","2023-01-03",
           "2023-01-04","2023-01-05"),
  CTR = c(2.3,2.7,2.0,2.4,2.6)
)
p1 <- ggplot(website1,
             aes(Date, PageViews, group=1)) +
  geom_line(color="blue") +
  geom_point(color="red", size=3) +
  labs(title="Daily Page Views") +
  theme_minimal()
p2 <- ggplot(website2,
             aes(Date, CTR, fill=CTR)) +
  geom_bar(stat="identity") +
  labs(title="Click Through Rate") +
  theme_minimal()

```

p1 + p2

OUTPUT:

